Digital QR Identity System

Final Prototype – Restructured Modules (Secure Model)

🔐 1️⃣ Authentication & Role Management Module

Objective

Provide secure access for internal users.

Users:

- Admin
- Faculty
- Student (for profile management only, not for QR viewing)

Functional Requirements

- Email + Password login
- Role-based dashboard
- JWT session handling
- Password encryption
- Logout
- OTP service integration (for QR access verification)

👨‍🎓 2️⃣ Student Management Module

Objective

Maintain official digital student records.

Functional Requirements

- Add student (manual + CSV)
- Edit student details
- Deactivate student
- Generate permanent QR (linked to PRN)
- View student profile (Admin only)

Student Profile Data

- PRN
- Name
- Year

- Department (ENTC)
- Email (for OTP)
- Contact details
- Document metadata (JSONB)

👩‍🏫 3️⃣ Faculty Management Module

Objective

Create public digital academic identity.

Functional Requirements

- Add faculty
- Edit faculty
- Generate QR (Employee ID based)
- Faculty uploads research
- Admin approval workflow
- Public profile page

Faculty QR Behavior

Scan → entc-college.com/faculty/{employee_id}

No login required.

Public information only.

📂 4️⃣ Document Management Module

Objective

Secure storage and controlled access of student documents.

Functional Requirements

- Admin uploads official documents
- Documents linked to PRN
- Document types:
    - Bonafide
    - Marksheets
    - Fee Receipts

    - ID Card
    - Certificates

Security Rules

- Documents NOT publicly accessible
- Stored securely
- Access controlled via OTP verification

🪪 5️⃣ QR Code Access Module (Core Logic)

This is the heart of the system.

🔹 Faculty QR Flow

1. Scan QR
2. Redirect to public faculty profile
3. No authentication required

🔹 Student QR Flow (Secure)

Step-by-Step Flow:

1. Scan student QR
   → Redirect to:
   entc-college.com/student/{PRN}
2. System shows:
    - Student basic info (Name + PRN only)
    - "Request Secure Access" button
3. OTP sent to registered email
4. User enters OTP
5. If verified:
    - Temporary session created (e.g., 10–15 mins)
    - Student documents unlocked
6. After expiry:
    - Access revoked automatically

Additional Controls

- OTP expiry (5 mins)
- Rate limit OTP attempts
- Audit log of access
- Admin can disable QR anytime

📋 6️⃣ Admin Dashboard Module

Objective

Central monitoring for ENTC.

Dashboard Features

- Total Students
- Total Faculty
- Documents uploaded
- Research approval pending
- QR scan analytics
- OTP verification logs
- Access activity logs

📊 7️⃣ Reports & Audit Logs Module

Objective

Maintain traceability and transparency.

Reports Include

- Student document access logs
- Faculty profile views
- Research approval history
- OTP request logs
- QR scan count per user

🔐 8️⃣ Security & Compliance Module

Controls Implemented

- Role-Based Access Control

- OTP-based document unlock
- Session timeout
- Token expiration
- HTTPS only
- Encrypted password storage
- Activity logging
- QR regeneration capability

🎯 **Final Stakeholder Positioning**

Faculty QR = Public Academic Identity

Student QR = Secure Document Gateway

This makes your system:

- Practical
- Secure
- Scalable
- Realistic
- Privacy compliant